

Chapter 142 — Laparoscopic and Robotic Surgery of the Kidney

- **First minimally invasive renal operation: laparoscopic nephrectomy, Clayman and colleagues, Washington University, St Louis, 1991, in an 85-year-old woman.**
- Since then applied to **every surgical renal pathology.**
- Against open surgery: **equivalent long-term outcomes, faster recovery, no additional cost.**
- **Robotic technology drove widespread adoption** of minimally invasive renal surgery.
- Consistent advantages over open surgery: **less pain and blood loss, shorter convalescence, better cosmesis.**

PATIENT EVALUATION AND PREPARATION

- **Same principles as general minimally invasive surgery (MIS) and the equivalent open operation** (Chapters 11, 12 and 18).
- **History and examination** anticipate intraoperative problems; **body habitus** dictates positioning and access.
- **Prior abdominal or retroperitoneal surgery** alters trocar placement and the choice of transperitoneal versus extraperitoneal access.
- With experience, **several former absolute contraindications are now relative**; careful selection matters most **during the learning curve.**

Anaesthetic Considerations for Laparoscopic and Robotic Renal Surgery

- **General anaesthesia is mandatory**; cardiopulmonary function must tolerate it.
- **Pneumoperitoneum compromises ventilation, venous return and cardiac output.**
- **High-risk comorbidities: chronic obstructive pulmonary disease (COPD), intracranial pathology, cardiac disease** — each poses haemodynamic problems with insufflation.
- **Severe COPD may be unable to compensate for pneumoperitoneum-induced hypercarbia** → options: lower insufflation pressure → **helium insufflation** → **conversion to open surgery.**
- **AirSeal Intelligent Flow System** (valveless trocar): continuous **low-pressure, high-flow pneumoperitoneum** → lower carbon dioxide (CO₂) consumption → potentially better tolerance of laparoscopy.
- Robotic surgery:
 - **Dedicated robotic theatre** to accommodate robot, tower and console.
 - Robot heavy and bulky with a **wide external arc of arm movement** → **protect head, shoulders and arms from collision.**
 - **Docking and undocking delay patient assessment and management in an emergency.**
 - Surgeon at the console is **less likely to notice the patient slipping** → **close communication between anaesthetist, bedside assistant and surgeon** is essential.

Considerations in Obese Patients

- Difficulties: **access; anatomy hidden or distorted by fat; restricted trocar and instrument range; longer instruments; higher pneumoperitoneum pressures.**
- With experience and technical modification, **safe in obese patients.**

- **Body mass index (BMI) is a poor predictor of difficulty:** it separates neither fat from muscle nor visceral from subcutaneous fat.
- **Visceral adipose tissue is the better index of operative difficulty,** particularly when BMI is normal.

Considerations in Elderly Patients

- **Benefits of MIS apply irrespective of age.**
- Particularly valuable in the elderly: **less pain and opioid use → early mobilisation → shorter stay.**
- **Safe and valid in carefully selected elderly patients.**

Concept — What the patient must tolerate is the **pneumoperitoneum**, not the renal operation; cardiopulmonary reserve, not age or BMI, therefore governs fitness for minimally invasive renal surgery.

SURGICAL APPROACHES AND OBTAINING ACCESS

- Categories, each laparoscopic or robotic: **transperitoneal; retroperitoneal; hand-assisted; single-site; natural orifice transluminal endoscopic surgery (NOTES).**
- Choice depends on **resources, surgeon experience, and patient and disease characteristics.**
- **Transperitoneal is the most widely used; retroperitoneal remains viable for selected patients and pathologies,** and demands familiarity with landmarks to create a working space **without breaching the peritoneum.**

COMPARISON — TRANSPERITONEAL VERSUS RETROPERITONEAL ACCESS

	Transperitoneal	Retroperitoneal
Working space	Large; many trocar and instrument options	Limited → poor trocar spacing, reduced triangulation
Landmarks	Familiar, easily identified	Less familiar
Bowel	Mobilisation needed; adhesiolysis after prior surgery or inflammation	Peritoneum not entered; no mobilisation of spleen, liver or colon
Hilum	Reached after colon reflection	Direct access, especially posterior aspect
Optics	Unremarkable	Dissection close to the lens → frequent smudging
Oncological	—	Theoretically saves peritoneal cavity from tumour spillage
Preferred for	Most renal surgery; ease of suturing	Selected partial nephrectomy, pyeloplasty, cyst marsupialisation, renal biopsy; extensive adhesions, peritoneal dialysis, pregnancy, morbid obesity
Outcome in selected cases	Reference	Longer operation, shorter stay, earlier recovery; comparable perioperative and oncological outcomes

- **What the table is for:** choosing access — retroperitoneal trades working space for directness and an untouched peritoneum.
- **Trap:** retroperitoneal is **slower yet discharges sooner;** and it is the access of choice where the peritoneum must be preserved or avoided (peritoneal dialysis, dense adhesions).

Transperitoneal Approach

- **Most traditional and widely used** approach; advantages and disadvantages as in the table above.

Patient Positioning

- **Sequence:** intravenous access → induction → endotracheal intubation → bladder catheter → sequential compression stockings → modified flank position.
- **Modified flank:** operative side up 30–45°; back supported on rolled blankets or gel rolls.
- **Lower arm on a padded armrest, upper arm flexed at the elbow across the chest** (or on pillows); head and cervical spine on towels, blankets or foam donuts; **pillows beneath the knees.**
- **Pad all pressure points; secure with wide cloth or silk tape** so the table can be tilted; **rotate the bed to each side to test stability.**
- Theatre arranged so the **whole team can see the procedure** (left nephrectomy: two monitors; scrub technician placed for instrument exchange).
- **Prepare and drape the entire abdomen and flank** in case of conversion.

Trocar Placement in Transperitoneal Laparoscopic Surgery

- Pneumoperitoneum by **Veress needle or Hasson (open) technique** → **three to five trocars**; configuration adapted to surgeon preference.
- Template:
 - **10-mm camera port at the umbilicus.**
 - **5- or 10-mm midline port, about 2 cm below the xiphoid.**
 - **12-mm port, anterior axillary line at the level of the umbilicus** — sutures, clamps, staplers.
- **Short patient:** 12-mm port moved to the **midline, halfway between umbilicus and pubis.**
- **Obese patient:** shift all trocars laterally.
- Accessory ports: **low midline 10/12 mm for colon retraction; 5-mm subxiphoid for liver retraction.**

Figure notes —

- Fig. 142.3 alternative template: **12 mm lateral to the rectus at umbilical level, 10 mm at the umbilicus, 5 mm midline between umbilicus and xiphoid**; accessory subcostal, subxiphoid and low midline sites.
- Fig. 142.4 retraction: **left side** — blunt instrument via a 5/10-mm trocar above the symphysis, or instrument and retractor via a subcostal incision; **right side** — liver and bowel via a 3- or 5-mm midline trocar, with an optional 10-mm lower midline trocar freeing both working hands.

Transperitoneal Robotic Approach

- Allows surgeons with **limited laparoscopic skills** to perform MIS.
- Advantages: **three-dimensional (3D) view, more degrees of instrument motion, motion scaling, tremor elimination, ergonomic seated posture, wrist articulation easing suturing.**
- Positioning and port placement mirror laparoscopy.
- **Three-arm configuration:** camera periumbilical; two robotic ports in the anterior axillary line — **one above the iliac crest, one subcostal**; arm spacing 8 cm (Si) or 6 cm (Xi); **12-mm assistant port in the low midline** for sutures, clamps, staplers, suction, retraction.
- **Linear configuration:** all 8-mm ports along the ipsilateral lateral rectus sheath; camera just cranial to the umbilicus; one port 6–7 cm cranial, two caudal about 6–7 cm apart; assistant port midline between cranial and camera ports.
- **Whatever the template, ports form a broad-based triangle with the target anatomy.**
- **No stapler anticipated (e.g. pyeloplasty) → 5-mm assistant port suffices; additional 5- or 12-mm subxiphoid port for liver retraction.**

- **Short patients: risk of arm collision; obese patients: lateral shift.**

Concept — Every port template is the **same triangle** — camera at the apex, working ports at the base, target at the far point; body habitus only moves or compresses the triangle.

Retroperitoneal Approach

- **Resembles open flank surgery; peritoneal cavity not violated.** Indications, advantages, limitations and outcomes as in the table above.

Patient Positioning and Trocar Placement in Retroperitoneal Laparoscopy

- **Full flank; table flexion** widens the rib–iliac crest gap; **axillary roll** protects the lower arm; pad and secure for tilting.
- **Surgeon stands behind the patient.**
- **Initial incision: transverse, posterior axillary line, halfway between the tip of the 12th rib and the iliac crest.**
- Creating the space — options:
 - **Visual obturator + 12-mm trocar + laparoscope** → enter under vision → insufflate → sweep with the scope.
 - **Blunt split of flank muscles** → **lumbodorsal fascia divided** → **finger dissection** between posterior abdominal wall and kidney; correct position = **smooth psoas and lower pole within Gerota's fascia felt.**
 - **Balloon dilator** displaces fat and peritoneum anteromedially; **home-made from two fingers of a size 8–9 glove** → removed → **blunt-tip trocar with cuff cinched to skin** prevents gas leak.
 - **0° lens through a visual obturator; characteristic yellow retroperitoneal fat confirms the correct plane** → insufflate → develop with scope, working-channel Kittner or suction; peritoneum pushed medially.
- **Trap: balloon must lie wholly within the retroperitoneum** — inflation within muscle → **postoperative flank hernia.**
- Plane errors (pair): **too anterior** → **peritoneal breach or colon injury**; **too posterior** → **bleeding from quadratus lumborum or psoas.**
- Then orientate → **5-mm trocar just off the 12th rib tip; 12-mm trocar posterior and superior to the camera, both under vision.**
- **Bail-out: convert to transperitoneal by opening the peritoneum under direct vision.**

Figure notes —

- Fig. 142.7 variant: full lateral, hips flexed, kidney rest elevated; **15-mm incision 2 cm below the 12th rib tip**, between rib and anterior superior iliac spine; glove-finger balloon tied with silk over a red rubber catheter and **filled with 600–800 mL saline**; balloon-and-collar blunt-tip trocar needs no sutures and rotates 360°.

Patient Positioning and Trocar Placement in Retroperitoneal Robotic Surgery

- Space developed as for retroperitoneal laparoscopy.
- **8-mm camera port, midaxillary line, 2 cm above the iliac crest**; second and third ports in the **posterior and anterior axillary lines, 1 cm cephalad**; **assistant port 4 cm cephalad to the third arm, anterior axillary line.**
- **Robot docked over the patient's head**, not from the side → can dock perpendicular or parallel to the bed.

Hand-Assisted Laparoscopy

- Hand-assisted laparoscopy (HAL): **first HAL nephrectomy in the pig 1996; human 1998.**
- Combines open advantages — **tactile feedback, manual retraction, haemostasis** — with laparoscopic **visualisation and small incisions.**
- Uses: adjunct for surgeons with **limited laparoscopic training; popular in donor nephrectomy; hand port added when a laparoscopic case becomes difficult.**
- Suits the novice where **dense perirenal scarring** is expected, e.g. **xanthogranulomatous pyelonephritis (XGP)**; aids intracorporeal suturing and knot tying.
- **Emergency role: extend a port site → hand port → finger pressure on the bleeding vessel** while open conversion is prepared — keep in the armamentarium.

Patient Positioning and Trocar Placement in Hand-Assisted Laparoscopic Renal Surgery

- Positioned as for transperitoneal surgery.
- **Sequence: hand-port incision through skin, fascia and peritoneum → hand port placed → pneumoperitoneum → further trocars under vision → lubricated hand inserted; laparotomy sponge optional for retraction and haemostasis.**
- Site depends on **side, habitus, handedness, preference; incision not too large** (gas leak, loss of pneumoperitoneum); hand must not obstruct the laparoscope.
- **Forearm pressure 30–100 mmHg** → tingling, numbness and pain in hand and forearm.
- **Specimen extracted in an entrapment bag** through the hand port; **irrigate copiously before closure** (infection).

HAND-PORT PLACEMENT BY HANDEDNESS AND SIDE (FIGS. 142.9 AND 142.10)

Surgeon / kidney	Hand port (hand)	Working instrument	Camera	Additional
Right-handed / right	Right lower quadrant (left hand)	Umbilical trocar	Midline, several cm above umbilicus	Subcostal liver/bowel retractor — liver retraction usually needed
Right-handed / left	Periumbilical (left hand)	Subcostal margin, just medial to the nipple line	Several cm lateral to the edge of the hand device	Most lateral trocar
Left-handed / right	Periumbilical (right hand)	Lateral to rectus, at or just below umbilicus	Anterior axillary line	Subcostal liver retraction
Left-handed / left	Left lower quadrant (right hand)	Umbilical trocar	Midway umbilicus–xiphoid	Fourth trocar, subcostal margin

- **Pattern: the non-dominant hand goes in;** when the operative side matches the surgeon's dominant side, the hand port sits in the **ipsilateral lower quadrant**, otherwise **periumbilical.**
- **Trap: camera is positioned relative to the edge of the hand device, not the edge of the incision.**

Single-Site Minimally Invasive Renal Surgery

- Aim: **further reduce morbidity** by consolidating ports into **one skin incision** — **laparoendoscopic single-site surgery (LESS)** — or **through a natural hollow viscus (NOTES).**
- **LESS: entire spectrum of renal procedures performed; pure NOTES: very few.**
- **Hybrid NOTES: standard laparoscopy with the vagina as access and extraction route** in nephrectomy.
- NOTES adoption limited by **no clear therapeutic benefit, safety concerns, restrictive selection.**
- LESS: **longer operation, shorter stay, less analgesia;** held back by **lack of triangulation, optical limitation, technical complexity.**

Patient Positioning and Trocar Placement in Laparoscopic Single-Site Surgery

- Access at the **umbilicus or below the waistline** to hide the scar; **modified flank** (transperitoneal) or **full flank** (retroperitoneal).
- Either **several low-profile trocars close together in one small extraction incision** or a **purpose-built multichannel port**.
- Conventional instruments usable; **flexible or articulating instruments add degrees of motion**.

MULTICHANNEL SINGLE-SITE ACCESS DEVICES

Device	Channels
TriPort 15	Two 5-mm + one 15-mm
TriPort+	Three 5-mm + one 10-mm
da Vinci SP entry guide	Three articulating 6-mm instruments + one flexible articulating camera through a single 27-mm guide

Figure notes —

- Fig. 142.11: LESS with three low-profile trocars in one extraction incision, flexible scope and instruments (from Tracy et al., 2008); LESS donor nephrectomy with a multichannel device plus a **2-mm instrument** for retraction, hilar dissection and extraction.

Single-Port Robotic Surgery

- Robotic surgery (2000) revived single-site surgery — aims: **less trauma, shorter stay, faster recovery**.
- **First single-port (SP) robotic urological surgery 2009**: standard robot, **12-mm scope + two robotic ports through a multichannel port in a 2-cm umbilical incision**; successful, but technically difficult.
- **da Vinci SP, 2018**: purpose-designed; **27-mm entry guide** (see table); **on-screen navigator** shows each instrument's position in real time to prevent collision.
- Limited availability; feasibility, safety and promising outcomes reported.
- **Assistant port** for devices the platform lacks (staplers, suction).

Patient Positioning and Trocar Placement in Single-Port Robotic Renal Surgeries

- **Flank 60°, flush with the bed edge; bed elevated** so the arm clears the floor and the assistant can exchange instruments.
- **10–25 cm between cannula tip and target** — room to deploy elbow and wrist joints; avoids external collision.
- **Sequence: 3-cm periumbilical incision → entry guide → insufflation at 10–12 mmHg**.
- Assistance: separate **AirSeal trocar**, or a **GelPort** housing insufflation and entry guide and giving distance to the target; **mini-GelPort or SP Access Kit** prevents gas leak and accommodates entry guide, AirSeal port and flexible suction.
- **Remote-operated suction irrigation (ROSI)** device steered by a robotic arm gives constant suction and irrigation.
- If assistance through the GelPort fails → **extra 5- or 10-mm lower-quadrant assistant port**.
- **Mini-Pfannenstiel alternative**: cosmesis, less pain → **opioid-free and same-day surgery pathways**.

Single-Port Retroperitoneal Access and Trocar Placement for Renal Surgeries

- **Narrow profile suits the confined retroperitoneum**; technique mimics multiport.
- **Sequence: full flank → 2.5–3-cm transverse incision 2 cm anterior and inferior to the 12th rib tip → split flank muscles → thoracolumbar fascia → enter retroperitoneum, sweep peritoneum medially → balloon dilatation → 25-mm GelPort or Access Kit → entry guide**.
- Instrument clock: **camera 12 o'clock; monopolar scissors 3 o'clock; Maryland forceps 9 o'clock**.

COMPARISON — MINIMALLY INVASIVE ACCESS PLATFORMS

Platform	Access	Gains	Costs
Transperitoneal laparoscopy	3–5 trocars	Space, landmarks	Bowel mobilisation, adhesiolysis
Robotic multiport	3–4 arms + assistant	3D, wristed instruments, tremor-free, ergonomic	Bulk, docking delays emergency access
Retroperitoneal	Flank, space created	Direct hilum, peritoneum spared	Small space, unfamiliar landmarks
Hand-assisted	Hand port + trocars	Tactile feedback, rapid haemorrhage control	Larger incision, forearm compression, more wound complications
LESS	One incision	Cosmesis, less pain	No triangulation , instrument clashing
NOTES	Natural orifice (hybrid: vaginal)	No abdominal wall scar	No proven benefit, safety concerns
SP robotic	One 25–27-mm guide	Shorter stay, less opioid	Limited availability

- **Pattern:** each step toward fewer incisions buys cosmesis and pain relief at the price of working space or triangulation.

Concept — Every approach trades **working space against access trauma**: transperitoneal maximises space, retroperitoneal gives directness, single-site minimises incisions at the cost of triangulation, and the hand port buys back tactile control at the cost of a larger wound.

SIMPLE NEPHRECTOMY

- Indications — **benign renal disease**: chronic pyelonephritis; obstructive or reflux nephropathy; renal tuberculosis; multicystic dysplastic kidney; renovascular hypertension; acquired renal cystic disease in dialysis patients; nephrosclerosis; **symptomatic autosomal dominant polycystic kidney disease (ADPKD)**; post-transplant hypertension.
- **Chronic refractory pyelonephritis, including XGP, can be managed minimally invasively**, but dense perinephric adhesions and lost planes make **simple nephrectomy at times harder than radical**, with a **higher conversion rate**.
- **Predictors of difficulty**: previous pyonephrosis and an indwelling nephrostomy tube.
- **Stone disease carries the highest intraoperative and postoperative complication rates** (national database).
- Rescue manoeuvres: **hand assistance or subcapsular nephrectomy**.

Procedure

- Safe **retroperitoneally or transperitoneally, laparoscopically or robotically**; steps similar to radical nephrectomy.

COMPARISON — SIMPLE VERSUS RADICAL NEPHRECTOMY

	Simple	Radical
Gerota's fascia and perirenal fat	May be left in situ → smaller specimen	Removed intact with the specimen
Typical pathology	Benign, often inflammatory	Renal cell carcinoma
Difficulty	Can exceed radical (adhesions, lost planes)	Planes usually preserved

	Simple	Radical
Conversion	Higher	Lower

Reflection of the Colon

• Left side:

- **Incise peritoneum along the white line of Toldt** from below the lower pole to above the upper pole, lateral to the spleen (endoshears, bipolar or ultrasonic energy); **avoid the diaphragm**; extend inferiorly as needed.
- **Leave the lateral renal attachments intact at this stage** — otherwise the kidney rolls over and obscures the hilum.
- **Divide the lienocolic ligament** → spleen, pancreas and colon fall medially; divide thin colorenal attachments → retract colon medially.
- **Key plane: brighter-yellow mesenteric fat versus pale-yellow retroperitoneal (Gerota's) fat** — the avascular plane between them avoids colonic injury and incision of mesocolon or Gerota's fascia.
- **Mesocolic defect → repair** (internal hernia).
- Paddle retractor via an extra low midline trocar retracts colon, pancreas and spleen.

• Right side:

- Liver retractor via the anterior axillary line or a high midline port to expose the upper pole.
- Medial traction on the colon reveals colorenal attachments → divide.
- **Cold sharp dissection around the duodenum** — avoids thermal injury.
- **Kocher manoeuvre** (lateral duodenal mobilisation) → exposes medial kidney, hilar connective tissue and **inferior vena cava (IVC)**.

Dissection of the Ureter

- **Sequence: psoas muscle and tendon below the lower pole → follow medially → gonadal vessels met first → ureter just posterior to them → elevate ureter, sweep gonadal vessels medially → follow up to the lower pole.**
- **Peristalsis distinguishes the ureter** from adjacent vessels.
- **Do not divide the ureter yet** — it is the handle for lifting the kidney.
- Sweep tissue behind ureter and lower pole anteriorly to bare the psoas; **stay above the psoas fascia** → protects cutaneous nerves → avoids **thigh numbness, pain, paraesthesia**.
- **Right side: tearing the gonadal vein at its insertion into the IVC during elevation causes significant bleeding.**

Identification of the Renal Hilum

- Exposure: colon and small bowel medial (gravity or low midline retractor) + **anterolateral lift of the kidney out of the renal fossa → hilum on stretch**.
- Lateral grasper under ureter and kidney until it abuts the sidewall; **never into the parenchyma**.
- **Layer-by-layer blunt dissection until the renal vein appears**; gonadal, lumbar and accessory veins occluded and divided as needed.
- **No clips in the anticipated staple line** → stapler misfire.

Securing the Renal Blood Vessels

- **Renal artery usually lies posterior to the vein**; **preoperative imaging** maps hilar anatomy.
- Blunt and sharp dissection: irrigator-aspirator tip, hook electrode, scissors, forceps.

- **Sequence: artery ligated and divided first → then vein**, with an endoscopic vascular (gastrointestinal anastomosis, GIA) stapler; **stapler must cover the whole vessel**.
- Left renal vein stapled **lateral to the adrenal vein**; any clips on gonadal or adrenal vessels **excluded from the stapler jaws**.
- **En bloc hilar stapling is a safe alternative** (see Radical Nephrectomy).
- Bleeding along the staple line → **reinforcing clips**.
- **Clips and staplers have equivalent safety** — device failure, severe haemorrhage, conversion, mortality (meta-analysis, donor nephrectomy).
- **Polymer clips safe when properly placed: 2-mm separation between the two clips, a few millimetres from the cut end of the artery**.

Dissection of the Upper Pole

- After hilar division, dissection continues **posteriorly and superiorly** to the upper pole.
- **Adrenal preservation**: incise Gerota's fascia anteriorly just above the hilum → peel perinephric fat circumferentially off the upper pole (LigaSure laparoscopically; vessel sealer robotically).
- **Then clip and divide ureter → divide lateral attachments → rotate kidney anteriorly over the liver (right) or spleen (left) → divide uppermost attachments under vision**.
- **Extreme fibrosis → subcapsular nephrectomy once artery and vein are controlled**.
- **Avoid diaphragmatic injury**.

Organ Entrapment and Extraction

- Specimen in an **extraction bag → intact through an extended trocar site or Pfannenstiel incision, or piecemeal after morcellation** (ring forceps, Kocher clamp) **under endoscopic vision** to avoid bag disruption.
- Fig. 142.20: camera moved to the lateral port and Endo Catch passed through the umbilical site; for Pfannenstiel extraction the Endo Catch enters through a **separate 10-mm suprapubic incision**. Morcellation technique: see Specimen Extraction under Partial Nephrectomy.

Postoperative Management

- **Diet as tolerated; urethral catheter out once comfortably ambulant; activity as tolerated**.
- **No heavy lifting for 4–6 weeks** where there is an extraction incision.

Results

- Laparoscopic versus open simple nephrectomy: **less pain and blood loss, shorter stay, quicker return to activity; comparable complications**.
- **Retroperitoneal and transperitoneal equivalent** in experienced hands — operative time, blood loss, stay.

Concept — Simple nephrectomy for inflammatory disease is paradoxically **harder than radical nephrectomy for tumour**, because inflammation obliterates the tissue planes that tumour usually leaves intact.

SURGERY FOR RENAL CYSTIC DISEASE

- **Renal cysts are common but rarely need surgery**.
- Indications: **pain, infection, obstruction, concern for malignancy** (Fig. 142.22: peripelvic cysts on delayed-phase contrast computed tomography [CT] in left flank pain).
- **Sequence: percutaneous aspiration first — tests whether the cyst causes the symptoms → recurrence → decortication, marsupialisation or unroofing**.

- **Laparoscopic treatment achieves better symptomatic and radiological success than percutaneous aspiration, with similar complications.**
- **Multiple and bilateral cysts can be treated in one operation.**
- **ADPKD: laparoscopic cyst decortication effective for cyst-related pain; end-stage renal disease with enlarged, symptomatic or infected kidneys → bilateral synchronous laparoscopic nephrectomy; intractable ADPKD pain → laparoscopic renal denervation (both efferent and afferent sympathetic fibres ablated).**
- **Cyst suspicious for malignancy → explore, biopsy, and treat by cryoablation, enucleation, or partial or radical nephrectomy.**

Procedure

- Mobilise → incise Gerota's fascia → expose capsule → identify cyst (**intraoperative ultrasound helps**); **hilar control usually unnecessary.**
- **Sequence: dissect cyst wall free before draining → aspirate (laparoscopic needle) → grasp and excise wall along the cyst–parenchyma junction → evacuate fluid by suction → biopsy suspicious areas (5-mm biopsy forceps) → fulgurate the base (electrocautery or argon-beam coagulator).**
- **Ablating the base risks inadvertent or occult entry into the collecting system; packing with haemostatic agents or suturing may be needed.**
- **Collecting system entered → close it and leave a drain.**
- **Malignancy found → extirpation or cryoablation of the remaining lesion or kidney.**
- Drain technique (Fig. 142.24): hemostat through a side stab incision → drain passed via a trocar to the hemostat → **colon brought back over the kidney and fixed to the sidewall to "reperitonealise" kidney and drain.**

Results

- **Highly effective:** symptom improvement in the great majority, high radiological resolution, durable pain relief; **major complications rare.**

Concept — *Aspiration is **diagnostic before it is therapeutic**: it proves the cyst is the pain generator before decortication is committed to.*

RENAL BIOPSY FOR MEDICAL RENAL DISEASE

- Central to diagnosis and management of medical renal disease; **default is percutaneous biopsy under CT or ultrasound guidance.**
- **Laparoscopic biopsy preferred for: failed percutaneous biopsy; anatomical abnormality (multiple cysts, aberrant vessels); bleeding diathesis; morbid obesity; solitary kidney; patient choice or non-compliance with percutaneous biopsy.**
- **Adequate tissue and excellent haemostasis in almost all cases; more glomeruli than percutaneous biopsy (matched-pair analysis).**

Procedure

- **Retroperitoneal is more efficient than transperitoneal; full flank; two-port technique usual (LESS also described).**
- **10-mm trocar above the iliac crest, posterior axillary line; 5-mm trocar at the same level, anterior axillary line.**
- **Ultrasound localises the kidney in abundant retroperitoneal and perinephric fat.**

- **5-mm two-tooth biopsy forceps:** two or three cortical samples from the lower pole after opening Gerota's fascia; or **16-gauge Tru-Cut needle** passed percutaneously under vision for deeper tissue.
- Haemostasis: **manual compression, electrocautery, topical collagen mesh, argon-beam coagulator.**

Results

- **Short operation, negligible blood loss, about one day in hospital, rare complications, diagnostic in all cases.**

Concept — *Laparoscopic biopsy exists for the patients in whom a **blind percutaneous pass is unsafe or has failed** — those who bleed, those with one kidney, those who cannot be reached.*

NEPHROPEXY

- **Surgical fixation of a mobile (ptotic) kidney.**
- **Nephroptosis:** descent >5 cm, or two vertebral bodies, on intravenous urography from supine to erect (Fig. 142.26, from El-Moula et al., 2008).
- **Usually right-sided; young, slim women; asymptomatic in 20% of healthy women.**
- **Operate only if: persistent localised pain or haematuria; restricted flow on colour Doppler; impaired urinary drainage on imaging.**

Procedure

- **Incise posterior peritoneum** → expose psoas and quadratus lumborum fascia → strip Gerota's fascia to the capsule → **complete nephrolysis** (free from peritoneum and colon) for full mobility.
- **Suture fixation (commonest): interrupted non-absorbable 2-0 polypropylene (Prolene),** lateral and posterior capsule to underlying muscle fascia; optional anterior capsule to parietal peritoneum.
- **Avoid the iliohypogastric and genitofemoral nerves** during suturing.
- **Sliding-clip tightening:** sutures placed loosely, ends joined by a clip, each clip slid down to tighten once all are placed.
- **Alternative sling: tension-free tape or polyglactin/polypropylene mesh** — needle below lower pole → through abdominal wall → tape over lateral margin → through wall → tightened until the kidney is fixed to the dorsal wall.

Results

- **High rates of pain relief, radiological correction and functional improvement on nuclear renography; no major complications; high satisfaction.**

Concept — *Nephroptosis is **common and usually harmless**, so the operative trigger is never the descent itself but its proven consequence — pain, ischaemia or obstruction.*

CALYCEAL DIVERTICULECTOMY

- **Usually asymptomatic and incidental.**
- **Intervene if: infection, haematuria, pain, hypertension, declining renal function.**
- **Mimics a simple cyst on ultrasound** → consider a diverticulum before treating a large cyst close to the collecting system; intravenous urography (IVU) or cross-sectional imaging with a delayed contrast phase clarifies.
- **Options: unroofing, excision, ablation, obliteration, partial nephrectomy.**
- **Principles: identify cavity and os** → remove stones → fulgurate the cavity → widen the infundibulum.

- **Endoscopic treatment first (Chapter 97); laparoscopic or robotic when endoscopy fails or is unsuitable.**

Procedure

- Expose capsule → **parenchymal divot overlies the diverticulum** → incise with electrocautery → cavity.
- **Hilar control only if partial excision is planned.**
- Endophytic or small: **intraoperative ultrasound and/or dye injected through a preplaced externalised ureteric catheter** locate diverticulum and os.
- **Sequence: stones removed into a bag → cavity obliterated (argon beam or monopolar) → infundibulum sutured closed → parenchyma reapproximated → perirenal fat into the defect (less recurrence) → drain.**

Results

- Small series only; **excellent results** reported.

CONTRAST — ENDOSCOPIC VERSUS LAPAROSCOPIC HANDLING OF THE DIVERTICULAR NECK

	Endoscopic	Laparoscopic / robotic
Infundibulum	Widened (drainage)	Sutured closed
Cavity	Fulgurated	Obliterated, packed with fat
Place	First line	Failed or unsuitable endoscopy

Concept — *The two routes pursue **opposite logics**: endoscopy keeps the diverticulum draining by widening its neck; laparoscopy abolishes it by closing the neck and obliterating the cavity.*

NEPHROLYSIS

- **Renal mobilisation with skeletonisation of the hilar vessels and upper ureter, and ligation of lymphatic channels.**
- Indications: select **haematuria/loin pain syndrome** and **chyluria**.
- **Persistent chyluria → loss of fat and protein → anaemia, hypoproteinaemia, fatigue.**
- Fig. 142.29 (from Eisner et al., 2009): **milky efflux from the left ureteric orifice**; retrograde pyelogram showing a **lymphorenal fistula**.

Procedure and Results

- **Lateralise by cystoscopy** — chylous urine seen emanating from the ureteric orifice.
- Laparoscopic or robotic, **transperitoneal or retroperitoneal**.
- Traditional: **full mobilisation, complete skeletonisation of the pedicle, removal of perinephric fat.**
- Alternative: lymphatic reflux occurs **only through hilar lymphatics** → **omit upper pole dissection.**
- **Omental wrap around the hilum** may add a barrier against recurrence.
- **Resolution visible immediately at on-table cystoscopy.**
- Retroperitoneoscopic pedicle ligation versus open (meta-analysis): **shorter operation, less blood loss, shorter stay, faster bowel recovery, fewer complications.**

RADICAL NEPHRECTOMY

- **Standard minimally invasive operation for renal cancer**; tumours **up to 25 cm** removed laparoscopically.

- Complex cancers — **IVC involvement, nodal disease, adjacent or distant organ involvement** — manageable minimally invasively.

Transperitoneal

Procedure

- Positioning and trocars as above; **extra trocars or a hand port for large tumours, IVC involvement or adjacent organ invasion.**
- **Identical to simple nephrectomy except that Gerota's fascia and fat are left intact on the specimen.**
- **Adrenalectomy en bloc:** ligate the renal vein medial to the adrenal vein take-off; ultrasonic or bipolar shears control the multiple adrenal arterial branches; inferior retraction of the specimen exposes the plane.
- **Adrenal-sparing:** open the fascia over the upper medial kidney; blunt instrument above the hilum gives anterior and inferior traction.
- **Regional lymph nodes; full hilar or retroperitoneal dissection for clinically node-positive (cN+) disease.**
- **Partial excision of diaphragm, pancreas, liver, spleen or bowel** feasible.

Results

- Minimally invasive versus open: **shorter stay, less blood loss, fewer transfusions, earlier oral intake, shorter convalescence, less analgesia, better cosmesis** — similar oncological outcome.
- Meta-analysis: **fewer postoperative complications** laparoscopically; **no difference in cancer-specific mortality, recurrence or intraoperative complications.**
- **5-year and 10-year recurrence-free, cancer-specific and overall survival equivalent or better** with laparoscopy in long-term series.
- National database, stage I–II renal cell carcinoma (RCC): minimally invasive approach — **shorter stay, fewer 30-day readmissions, overall survival advantage**, whether laparoscopic or robotic.

Retroperitoneal

Procedure

- **Sequence:** psoas muscle and tendon → develop space anterior to psoas → medial dissection reveals ureter → lift ureter and lower pole → vessels on stretch (arterial pulsation seen through tissue) → layer-by-layer dissection → right-angle dissector frees artery and vein → stapler or clips: artery, then vein → posterior dissection cephalad to upper pole → lateral, then anterior attachments.
- **Left side:** a lumbar vein usually must be ligated and divided to reach the hilum.
- **Continuous identification of landmarks** prevents disorientation.

Results

RETROPERITONEAL VERSUS TRANSPERITONEAL LAPAROSCOPIC RADICAL NEPHRECTOMY — EVIDENCE

Evidence	Verdict
Randomised trial (small)	No difference in blood loss, time, complications, stay
Randomised trial (larger)	No difference in blood loss, opioid use, stay, complications; retroperitoneal shorter operation
Randomised trial (third)	Retroperitoneal shorter operation, less pain and analgesia, shorter convalescence
Meta-analysis	Retroperitoneal shorter operation, fewer intraoperative complications; no difference in oral intake, analgesia, postoperative complications, transfusion, oncology

Evidence	Verdict
Tumours >7 cm	Retroperitoneal shorter operation, less blood loss, earlier diet; similar oncology

- **Bottom line:** comparable complications, analgesia, hospital course, convalescence and oncology; the one consistent retroperitoneal advantage is shorter operative time.

Hand Assisted

Procedure

- Positioning, hand port and trocars as described earlier; **non-dominant hand retracts and dissects bluntly throughout.**
- Toldt incised while the hand retracts the colon; **fingers give lateral traction on the kidney and medial traction on the bowel** → plane posterior to mesocolon, anterior to Gerota's fascia.
- Left: hand retracts spleen and pancreas while lienorenal attachments are divided. Right: hand lifts the liver to expose the upper pole.
- Psoas → ureter elevated; **gonadal vein elevated with the ureter on the left, reflected medially on the right.**
- Hand lifts kidney off psoas → ureter followed to hilum → **fingers pull kidney anterolaterally while thumb pushes bowel and mesentery medially** → skeletonise → fingers palpate the artery and guide stapler or clips → **artery, then vein.**
- Lateral and superior attachments divided with LigaSure or Harmonic scalpel under hand traction.
- **Never bluntly finger-dissect the adrenal off the upper pole** — bleeding.
- Rolled laparotomy sponge via the hand port aids retraction and haemostasis; hand-port incision allows rapid removal.
- **Hand-port metastases reported** → **specimen in a removal device before extraction**; irrigate and close.

Results

- **Recovery, morbidity and cost comparable to pure laparoscopy:** same operative time, complications, cost, stay, return to activity, pain.
- **Wound complications (infection, incisional hernia) slightly higher with HAL.**
- **Oncological outcomes comparable** to pure laparoscopy and open surgery.

Special Considerations

Large Tumours

- **Minimally invasive surgery is feasible and safe for most large renal masses.**
- Tumours >10 cm (registry): laparoscopic/robotic versus open — **shorter stay; similar readmission and mortality; conversion higher laparoscopic than robotic**; conversion adds about a day in hospital; minimally invasive use rising, open falling.
- Problems: **reduced working space; distorted landmarks** → **disorientation; hilum obscured by perihilar collaterals or by the kidney falling over it.**
- Aids: **continuous reference to preoperative imaging; intraoperative ultrasound; hand port or extra trocars** for lateral distraction and distributed retraction.
- **Approach individualised** to patient, pathology and surgeon proficiency.

En Bloc Hilar Vessel Stapling

- Former fear: **arteriovenous fistula** — **not borne out** (meta-analysis, about 2 years' follow-up, no fistula).
- **Same complications; operative time shortened by about three-quarters of an hour.**

Lymphadenectomy

- **Routine retroperitoneal lymph node dissection (LND) for non-metastatic renal cancer is controversial**, more so in the era of adjuvant therapy.
- **Staging and prognostic value; oncological benefit low.**
- Population cohort: **no overall survival benefit, even in clinically node-positive disease**; meta-analysis: **no change in metastasis, overall survival or recurrence**, and a risk factor for metastasis in high-risk patients.
- Limitations of the evidence: **no accurate imaging nodal reporting system; no standardised template.**

SUGGESTED LYMPHADENECTOMY TEMPLATES

Side	Nodes	Boundaries
Right	Paracaval, precaval, retrocaval, interaortocaval	Diaphragmatic crus → IVC bifurcation
Left	Para-aortic, pre-aortic; interaortocaval only if suspicious	Diaphragmatic crus → aortic bifurcation

- Dissect carefully and clip lymphatics → prevents lymphocele.
- American Urological Association (AUA), expert opinion: clinically concerning regional lymphadenopathy → dissect all clinically positive nodes.

Renal Vein and Caval Tumour Thrombus

- Laparoscopic and robotic surgery now used; **pure laparoscopic series limited but encouraging.**
- **Renal vein thrombus sequence: full mobilisation → early renal artery ligation → milk thrombus back toward the kidney (laparoscopic DeBakey, vessel loop or hand) → stapler on the renal vein beyond the thrombus, or laparoscopic Satinsky clamp isolating an IVC cuff → cuff excised en bloc with vein stump → cavotomy oversewn with Prolene.**
- **Early arterial ligation reduces collateral flow and may retract the venous thrombus** — as in open surgery.
- **Intraoperative ultrasound** defines the location and extent of the thrombus.

CONTRAST — SHORT VERSUS LONG CAVAL THROMBUS

Short caval thrombus	Long caval thrombus
Tangential Satinsky clamp on the IVC	Circumferential IVC dissection above and below → Rummel tourniquets or vascular clamps; control contralateral renal vein ; ligate lumbar veins → open IVC → excise thrombus → close with Prolene → flush with heparinised saline → restore flow

- **Robotic IVC thrombectomy** versus open (systematic review): **less blood loss, fewer transfusions, shorter stay, fewer complications; same progression-free and overall survival.**
- **Preoperative angioembolisation benefits the robotic (unlike the open) procedure:** decompresses venous collaterals, improves robotic efficiency, allows early left renal vein disconnection in left RCC.
- **Sequential clamping: infrarenal IVC → contralateral renal vein → suprarenal IVC.**
- **Level III thrombus: complete liver mobilisation + tourniquet on the suprahepatic infradiaphragmatic IVC above the thrombus.**
- Endovascular adjuncts: **retrograde intracaval Fogarty balloon; balloon catheter via the right internal jugular vein above the thrombus.** Longer thrombi: IVC cross-clamping and cavotomy.
- **Safe and feasible, but catastrophic potential → expert surgeons only.**

Cytoreductive Nephrectomy

- **Important element of multimodal therapy for metastatic RCC; survival benefit shown.**
- Laparoscopic versus open: **quicker convalescence, less blood loss, shorter stay, no more complications; earlier start of systemic therapy.**
- Survival: **mixed** — no difference at 1 year in an early series; longer survival with laparoscopy in a later matched series.

Concept — Across every radical nephrectomy variant, access changes **morbidity and recovery, never oncology**; the oncological principles — early arterial control, en bloc removal, intact extraction — are carried over from open surgery unchanged.

PARTIAL NEPHRECTOMY

- **Partial nephrectomy (PN), where feasible, is now the default for renal masses:** preserved function, less morbidity, equivalent oncology → paradigm shift from radical nephrectomy (RN).
- Minimally invasive versus open PN: **shorter operation, less blood loss, shorter stay.**

Indications

- **First laparoscopic PN 1993 (Winfield); retroperitoneal approach 1994.**
- Initially **small exophytic clinical T1a masses** (Fig. 142.33); now **large and complex tumours.**
- **RENAL nephrometry score:** Radius; Exophytic/endophytic; Nearness to the collecting system; Anterior/posterior; Location relative to the polar lines.
- Score predicts: **complications, warm ischaemia time, conversion to RN, non-neoplastic parenchymal loss, collecting-system repair or injury, operative time, stay, blood loss, recurrence.**

Clinical Stage T1b and Greater Tumours

- **No randomised trials of PN versus RN for large masses;** nephron-sparing surgery is feasible.
- PN: **better renal function; lower chronic kidney disease (CKD), recurrence, cancer-specific and all-cause mortality.**
- **cT1–2 tumours upstaged to pT3a after PN:** oncology not compromised, function possibly better.
- **>4 cm (cT1b+), laparoscopic PN versus RN: same margins and survival; two-stage CKD progression only after RN; far greater estimated glomerular filtration rate (eGFR) decline after RN; PN longer and bloodier but equal complications and oncology.**
- **pT1b: laparoscopic and robotic PN lower positive-margin rates than open;** robotic less blood loss, shorter ischaemia, fewer intraoperative complications than laparoscopic — in high-volume centres (**>50 PN per year**).
- **cT2a: robot-assisted PN (RAPN) versus minimally invasive RN — longer operation, more blood loss, but matched analysis: same oncology and morbidity; RN worse eGFR decline.**
- **Contrary signal: tumours >5 cm upstaged to pT3a recur more after PN than after RN.**

Concept — For large tumours the case for PN rests on **renal function, not oncology**; the one oncological caveat is the **>5 cm tumour upstaged to pT3a.**

Laparoscopic Heminephrectomy

- **Excision of 30% or more of the renal parenchyma.**
- Technical issues: **deeper resection; transection of large intraparenchymal vessels (clips or sutures); intentional pelvicalyceal entry, sometimes needing repair.**
- Versus matched PN: same blood loss, time, stay, complications, creatinine; **larger resection, more collecting-system repair, longer warm ischaemia.**

- **Robotic heminephrectomy** for complex extensive resection: preserves function; versus matched RN, comparable complications, recurrence and survival.

Central and Hilar Tumours

DEFINITIONS — CENTRAL VERSUS HILAR TUMOUR

Central	Hilar
Abutting or invading the central renal sinus fat and/or collecting system (Fig. 142.34)	In the renal hilum, in direct contact with the renal artery and/or vein

- Central tumours: deep infiltration → **intraoperative ultrasound, intentional collecting-system entry and repair, complex reconstruction — all within the ischaemia time limit.**
- Central versus peripheral: **same blood loss, complications, margins and creatinine; longer warm ischaemia** (and, variably, longer operation and stay).
- **Preoperative imaging review + intraoperative ultrasound** define anatomy and resection plane; **3D reconstruction of triphasic spiral CT** maps the vessels.
- Hilar robotic PN: **longer warm ischaemia, same outcomes**; robotic versus laparoscopic for hilar masses: **shorter warm ischaemia, same margins.**
- **Hilar tumour abutting critical structures → enucleation**; an opened pelvicalyceal system serves as a safe, identifiable resection plane.
- Renorrhaphy: **avoid major vessels at the tumour base; superficial bites over a bolster** rather than deep sutures.

Tumour in a Solitary Kidney

- Challenging whatever the approach.
- Robotic PN: **short warm ischaemia, high negative-margin rate, conversion to RN rare**; complications not uncommon.
- **Robotic versus open: same renal function, fewer complications.**

Multiple Tumours

- **Nephron-sparing surgery favoured for multifocal ipsilateral tumours because of the risk of contralateral recurrence.**
- **Bilateral laparoscopic PN: staged or single-setting**; single-setting successful in most selected cases.
- **Sequential tumour-specific selective segmental arterial clamping guided by 3D reconstruction** for multiple ipsilateral tumours.
- Multiple versus single: **longer operation and stay; same blood loss, complications and function.**

Other Indications

- **With adrenalectomy for upper-pole tumour; renal artery disease; anomalous kidney (horseshoe); obesity; hereditary RCC.**
- Goal in all: **remove tumour, spare normal parenchyma, minimise ischaemia and operative time, minimise complications.**

Procedure

- **Primary goal: complete excision with negative margins and adequate haemostasis, with attention to warm ischaemia time.**
- Requires **3D understanding of renal anatomy, real-time reading of visual cues, efficient intracorporeal suturing.**

COMPARISON — TRANSPERITONEAL VERSUS RETROPERITONEAL PARTIAL NEPHRECTOMY

	Transperitoneal	Retroperitoneal
Advantages	Large space, familiar landmarks, versatile instrument angles, easier suturing	No bowel mobilisation; direct access to posterior kidney and hilum
Costs	Bowel mobilisation	Smaller space, less familiar landmarks
Typical tumour	Anterior	Posterior
Laparoscopic outcomes	Reference	Shorter operation, less blood loss, shorter stay; same complications and conversion — subject to selection bias (smaller lesions, less collecting-system repair)
Robotic outcomes	Reference	Less blood loss, shorter stay; same time, ischaemia, conversion, margins; unchanged when stratified anterior versus posterior

- **Choice rests mainly on tumour location and surgeon experience;** also tumour size, number of tumours, **number of renal arteries**, visceral fat, prior abdominal surgery.
- **Trap:** retroperitoneal PN looks better in comparative series partly because it is **chosen for smaller, simpler tumours**.

Transperitoneal Laparoscopic Partial Nephrectomy

- Initial steps as for transperitoneal access to the kidney.

Retroperitoneal Laparoscopic Partial Nephrectomy

- Enter retroperitoneum → **identify psoas** → **lift kidney anteriorly off psoas to see arterial pulsation** → dissect hilum for **bulldog clamp** placement when needed.

Robotic-Assisted Laparoscopic Partial Nephrectomy

- **Robot eases intracorporeal resection and suturing** → **shorter learning curve** for those not skilled in pure laparoscopy → spread of minimally invasive PN.
- Large multi-institutional cohort: excellent perioperative, functional and oncological outcomes; **trifecta predicted by BMI, baseline eGFR, tumour size and complexity**.
- **Trifecta: negative margin + no major complication within 30 days + warm ischaemia ≤25 minutes**.
- Bedside assistant: **hilar clamping, suction and retraction, suture delivery and cutting, clip placement, intraoperative ultrasound**.

Tumour Localisation and Excision

- **Sequence:** inspection + intraoperative ultrasound (size, depth) → enter Gerota's fascia away from the lesion → expose capsule (subcapsular dissection if fat adherent) → score capsule circumferentially (monopolar scissors or hook) → clamp hilum → incise or bluntly fracture parenchyma beneath the score (cold shears) → grasper and suction countertraction → complete excision.
- Hilar clamp (Fig. 142.37): **en bloc laparoscopic Satinsky**, or separate bulldogs on artery and vein.

CONTRAST — STANDARD PARTIAL NEPHRECTOMY VERSUS TUMOUR ENUCLEATION

	Standard PN	Enucleation
Plane	Rim of normal parenchyma	Natural plane around the tumour pseudocapsule
Perioperative	Reference	Less blood loss and function loss, shorter operation and stay, fewer complications
Margins	Lower positive rate	Higher positive rate
Oncology	Same recurrence, cancer-specific and overall survival	

	Standard PN	Enucleation
Consider when	Default	Familial RCC, multifocal disease, anatomical or functional solitary kidney, severe renal dysfunction

Haemostasis

- **Suture renorrhaphy is the most reliable technique**, ± haemostatic or sealing agents or an oxidised cellulose (Surgicel) bolster.
- **Sliding-clip renorrhaphy**: locking clips slid down the suture to the capsule keep the parenchyma under tension (preplaced Lapra-Ty clip at the tail; optional pledgets prevent capsular tearing; argon beam to the bed first).
- **Large vessels in the bed sutured or clipped individually**; visible vessels clipped during excision.
- **Barbed suture as a horizontal mattress — no knot tying → shorter operating and ischaemia times.**
- **Single-layer versus double-layer closure: single layer gives better function and fewer pseudoaneurysms.**

TISSUE SEALANTS AND HAEMOSTATIC AGENTS

Class	Product
Gelatin matrix thrombin sealant	Floseal
Fibrin glue	Tisseel
Fibrin sealant	Evicel
Polyethylene glycol hydrogel	Coseal
Cyanoacrylate glue	Dermabond
Surgical adhesive	BioGlue

- Most high-volume urologists use them; **no randomised study shows a clear benefit for any agent.**
- **Sutureless renorrhaphy with haemostatic agents** in selected cases.

Collecting System Repair

- **Pelvicalyceal system violated in about 60% of laparoscopic PNs.**
- **Risk factors: larger, endophytic and solid tumours.**
- Repair: **running 3-0 or 2-0 polyglactin (Vicryl)** — selective interrupted or figure-of-eight sutures, or incorporated into the running inner-layer renorrhaphy; or close parenchyma and capsule over the defect without primary repair.
- **Either way, leave a drain** — perinephric urinoma.
- Repair versus no repair: **longer operation and warm ischaemia; no difference in function or complications, including fistula.**
- **Routine ureteric catheter does not prevent urinary fistula.**
- Fig. 142.40: defect identified with vessels occluded and edge lifted with the needle tip; **integrity checked by intravenous indigo carmine or retrograde instillation; argon-beam coagulation can destroy the repair suture.**

Renal Hypothermia

- Techniques: **surface ice slush; cold saline via retrograde ureteric catheter; intra-arterial cold perfusion.**
- **Feasible, but functional benefit unproven**; most tumours do not need prolonged warm ischaemia.

- **Consider in: solitary kidney, significant renal insufficiency, complex tumour, anticipated long ischaemia.**

Warm Ischaemia and Hilar Control

- **Bulldog clamps on the artery $\pm$ vein, or laparoscopic Satinsky for en bloc clamping.**
- **Artery-only versus artery-vein clamping: no difference** in surgical outcome or short-term function (meta-analysis).
- Traditionally, excision and renorrhaphy entirely on clamp; **historical safe limit 30 minutes.**
- "Every minute counts" **challenged: saved parenchymal volume predicts renal function better than warm ischaemia time; threshold for irreversible injury remains controversial.**
- Ischaemia-reducing modifications: **off-clamp PN, super-selective clamping, early unclamping before renorrhaphy is complete.**
- **Early unclamping may reduce renal artery pseudoaneurysm** after robotic PN — high-quality data lacking.

Off-clamp partial nephrectomy

- **Small, exophytic, non-infiltrating tumours;** with experience, even larger, deeper, central or hilar tumours.
- Off- versus on-clamp: **better early eGFR, slightly fewer complications, more blood loss; no eGFR difference beyond the early postoperative period.**
- Randomised trials (robotic): **no difference in eGFR or split function,** up to 2 years.

Selective arterial clamping

- Clamp **the segmental branch(es) supplying the tumour** → relatively bloodless field without global ischaemia.
- Versus hilar clamping: **better function, longer operation, more transfusion;** similar complications and stay.
- Aids: **duplex ultrasound; near-infrared fluorescence (NIRF) imaging.**
- **Randomised trial, super-selective RAPN with NIRF versus conventional RAPN with early unclamping: no difference in 6-month function — stopped for futility.**

Parenchymal compression

- Hand compression (hand-assisted) or laparoscopic parenchymal clamp; **feasible and safe for peripheral tumours.**

COMPARISON — HILAR CONTROL STRATEGIES IN PARTIAL NEPHRECTOMY

Strategy	Ischaemia	Best suited	Verdict
En bloc (Satinsky)	Global, arterial + venous	Any	Equivalent to artery-only
Artery-only bulldog	Global arterial	Any	Equivalent to artery-vein
Early unclamping	Shortened global	Any	Possibly fewer pseudoaneurysms
Selective / super-selective	Segmental only	Mapped segmental supply	No functional advantage in randomised trial
Off-clamp	None	Small, exophytic	No long-term eGFR difference; more blood loss
Parenchymal compression	Regional	Peripheral	Feasible, safe

Strategy	Ischaemia	Best suited	Verdict
Hypothermia	Cold, prolonged	Solitary kidney, long ischaemia	Benefit unproven

- **Pattern:** every strategy to shorten or localise ischaemia has failed to beat standard clamping on long-term function — consistent with **parenchymal volume, not ischaemia time, governing outcome**.

Concept — For function, **what is kept matters more than how long it is clamped**; clamping strategies have yielded less than their theory promised, whereas preserving parenchyma — enucleation, precise excision — carries the functional benefit.

Results

- PN versus RN: **equivalent oncology in properly selected patients**.
- **Long-term (5- and 10-year) recurrence-free and overall survival equivalent for laparoscopic and open PN in pT1**; metastasis predicted by **tumour size, absolute indication and comorbidity** — **not surgical approach**.
- Robotic PN in cT1 localised masses: **highest trifecta rate** (here warm ischaemia <20 minutes, no positive margin, no complication); **American Society of Anesthesiologists (ASA) score 3–4 and collecting-system involvement predict failure**.

COMPARISON — MINIMALLY INVASIVE VERSUS OPEN PARTIAL NEPHRECTOMY

Comparison	Advantage	No difference
Laparoscopic vs open	Shorter operation, less blood loss, shorter stay ; but in early large series more postoperative complications and longer warm ischaemia laparoscopically	Intraoperative complications, margins, 3-year oncological and functional outcome
Robotic vs laparoscopic	Fewer conversions (open and RN), shorter warm ischaemia, smaller eGFR change, shorter stay	Complications, time, blood loss, margins
Robotic vs open	Less blood loss, shorter stay, fewer complications	Warm ischaemia, margins, conversion, eGFR, cost

Positive Surgical Margins

- **Negative margins are always the goal**; positive margins occur in **0–7% of minimally invasive PN series**.
- **Recurrence risk doubled only in high-risk tumours (pT2–3 or nuclear grade 3–4)**; low-risk tumours (pT1, Fuhrman grade I–II) not affected.
- Other series: **disease-free survival unaffected by margin status**; **advanced clinical stage and pathological upstaging, not margin status, predict recurrence**.
- **Microscopic positive margin → careful surveillance, not immediate reintervention**, particularly in advanced stages.

Tumour Seeding and Port-Site Recurrence

- **Rare** — **about 0.1% of laparoscopic cancer operations**, most of them port-site recurrences.
- Potential factors: **tumour aggressiveness, immune status, local wound factors, surgical technique, tumour spillage**.
- **Minimally invasive PN does not increase atypical recurrence (propensity-matched)**.
- **Not contributory**: pneumoperitoneum, aerosolisation of tumour cells, insufflation gas, wound closure technique.
- Prevention: **sound technique, no spillage, extraction in an impermeable retrieval bag**.

Specimen Extraction

CONTRAST — INTACT EXTRACTION VERSUS MORCELLATION

	Intact	Morcellation
Incision	Extended port site or Pfannenstiel	Much smaller , better cosmesis
Operative time, pain, stay	No significant difference	
Pathology	Full staging	Histological type and grade usually preserved; size, capsular and renal vein involvement (RCC) and depth of invasion (urothelial carcinoma of the renal pelvis) may be lost
Seeding	—	Single port-site recurrence in a large series; considered safe with correct technique

- **Morcellation rules:** purpose-built sac; adequate draping; change gowns, gloves and instruments afterwards.
- **LapSac is impermeable to bacteria and tumour cells, even after morcellation.**
- **LapSac sequence:** moistened hydrophilic wire through every third hole → sac rolled from the bottom and passed through a 12-mm port → trocar replaced (wire and drawstrings outside) → specimen placed in sac held open by the wire → wire removed → drawstrings and neck brought out through the periumbilical incision, held tight → site enlarged by 1 cm → ring forceps, alternating bites on protruding tissue.
- **No deep forceps passes** (bowel may be caught); **maintain pneumoperitoneum and laparoscopic view** of the sac to prevent injury and perforation.
- Fig. 142.21: LapSac introduced through the lateral 10-mm site; lateral 5- or 3-mm port helps load the specimen; ring forceps or Kelly clamp for manual morcellation.

Local Recurrence and Repeat Minimally Invasive Surgery

- **Isolated local recurrence:** ipsilateral retroperitoneal nodes, renal fossa or adrenal — rarely recurrent venous thrombus — without distant metastasis.
- **Incidence about 2% after RN.**
- Literature mostly open (morbid); **small laparoscopic series feasible without conversion**; recurrence may appear **many years** after nephrectomy.
- **Surgical resection with curative intent + systemic therapy is superior to either alone**, even with metastatic disease; **large recurrence → consider intraoperative radiotherapy.**
- **Salvage robotic surgery after PN or ablation:** salvage RAPN (sRAPN) or salvage robot-assisted RN (sRARN); after prior PN most need sRARN; complexity may force RN; no major complications or conversions; good metastasis-free survival.
- **Recurrence in a prior resection bed is infiltrative, without a pseudocapsule → avascular planes hard to find.**
- RAPN for recurrence after ablation or PN: **low conversion, transfusion and major complication rates; eGFR largely preserved.**

Surgical Salvage After Failed Ablative Therapies

- **Minimally invasive PN and RN feasible after failed cryoablation or high-intensity focused ultrasound (HIFU).**

- **Perinephric fibrosis** → **lost tissue planes** → **collecting-system injury, possibly ureteric stent**; bleeding needing angiographic embolisation reported.

Concept — *Salvage surgery is hard for one reason: **the planes are gone** — fibrosis after ablation, infiltrative regrowth without pseudocapsule after PN.*

LAPAROSCOPIC ABLATIVE TECHNIQUES

- **Temperature extremes kill tumour cells**; goal: **complete destruction with minimal or no morbidity**.
- Versus PN: **less blood loss, shorter operation, less dissection, fewer complications**.
- **Commonest indications**: significant comorbidity, solitary kidney, hereditary RCC.
- Modalities: **cryoablation, radiofrequency ablation (RFA), microwave**.

CONTRAST — PERCUTANEOUS VERSUS LAPAROSCOPIC ABLATION

	Percutaneous	Laparoscopic
Status	Preferred for small renal masses (lower morbidity)	Reserved for highly selected patients
Advantages	Shorter stay, lower cost	Direct vision, renal mobilisation, retraction of adjacent structures
Cryoablation efficacy	More incomplete ablation	Less incomplete ablation, higher cancer-specific survival
Complications	Similar overall; more Clavien I	Similar overall; more Clavien III

Cryoablation

- Transperitoneal or retroperitoneal by tumour location; **no hilar clamping**.
- **Sequence**: mobilise → incise Gerota's → expose tumour (overlying fat may be excised) → biopsy (14- or 18-gauge needle) → cryoprobes placed percutaneously under laparoscopic and ultrasound guidance → two freeze-thaw cycles → probes removed with a gentle twist.
- Probes **parallel, in a triangular or quadrangular array with overlapping cryolesions**; number and spacing set by probe ablation geometry and tumour size.
- **Probe tips just beyond the deepest tumour margin**.
- **Iceball monitored in real time by ultrasound**; extends 0.5–1 cm beyond the tumour edge.
- **Iceball must not touch collecting system, ureter, renal vessels or adjacent organs**.
- Bleeding: pressure, or fibrin glue or Floseal.

Radiofrequency Ablation

- Transperitoneal or retroperitoneal; **no hilar control**.
- Probe into tumour → **tines deployed to cover tumour plus a 1-cm margin of normal tissue**.
- Mechanism: **frictional heating from ionic oscillation** under monopolar or bipolar high-frequency alternating current → **60–100°C** throughout the tumour.
- **Avoid >105°C — tissue vaporisation**.
- Ablation-zone size depends on **electrode length, diameter, surface area and temperature**.
- **No imaging monitors RFA in real time** → treatment guided by **direct temperature or electrical impedance**.

COMPARISON — CRYOABLATION VERSUS RADIOFREQUENCY ABLATION

	Cryoablation	RFA
Energy	Freezing	Frictional heat, 60–100°C (<105°C)
Margin	Iceball 0.5–1 cm beyond tumour	1 cm of normal tissue
Monitoring	Real-time ultrasound of iceball	None by imaging; temperature or impedance
Cycles	Two freeze–thaw	—
Hilar clamp	Not needed	
Usual route in series	Mostly laparoscopic	Mostly percutaneous
Outcome	Fewer retreatments, better local control, less metastatic progression (indirect comparison)	More retreatment

Results

- **Effective for small renal masses ≤ 3 cm;** laparoscopic ablation now **reserved for a highly selected group** because percutaneous techniques have advanced.
- Laparoscopic cryoablation for T1a: **low treatment failure; durable 5- and 10-year disease-free survival.**
- **Failure predictors:** larger size, endophytic lesion, hilar location; 4–7 cm masses recur more after cryoablation than after PN.
- **No randomised trials against PN.** Meta-analyses: ablation — **better perioperative outcomes but higher local progression;** metastatic progression higher in one analysis, similar in another.

Complications

- Laparoscopic versus percutaneous: **similar overall rate; Clavien I commoner percutaneously, Clavien III commoner laparoscopically.**
- Laparoscopic cryoablation: **modest overall complication rate, low major rate.**
- **Risk factors:** larger tumour, pre-existing heart disease, female sex.

Concept — *Ablation trades oncological certainty for low morbidity; its place is therefore the small (≤ 3 cm), exophytic tumour in a patient who cannot afford an operation — and for that patient the percutaneous route has displaced the laparoscopic.*

SINGLE-SITE MINIMALLY INVASIVE RENAL SURGERY

Laparoendoscopic Single-Site Surgery of the Kidney

- Used for pyeloplasty, simple, donor and radical nephrectomy, nephroureterectomy, PN, cyst decortication, cryoablation.

Renal Laparoendoscopic Single-Site Surgery

- Same steps as conventional laparoscopy; problems: **limited triangulation and instrument clustering.**
- Solutions: **flexible scopes, articulating instruments, offset optics, custom handpieces, instruments and scopes of varied length.**
- Multi-institutional experience: **an additional port often needed; conversion mostly to conventional laparoscopy, rarely to open.**
- Randomised trial versus laparoscopy: same time, blood loss, complications, stay; **less analgesia.** Donor nephrectomy meta-analysis: same stay, recovery, transfusion, conversion, analgesia; **lower pain scores.** Renal mass: **same perioperative outcomes and survival.**

- **Net: less pain, better cosmesis, no loss of perioperative or long-term outcome — but extensive laparoscopic skill required → high-volume centres, selected patients.**

Single-Port Robotic Renal Surgery

- Evolved from the 2009 exploration; **purpose-built platform, United States Food and Drug Administration (FDA) approval 2018**; not yet widely available; **safe and feasible transperitoneally and retroperitoneally.**
- SP-RAPN versus multiport (MP) RAPN: **shorter stay, less opioid**; same operative time, blood loss, ischaemia, margins, complications, transfusion.
- Systematic review (PN, pyeloplasty, adrenalectomy): SP PN — **longer warm ischaemia, same total operative time**; same complications and oncology; **less pain, same-day discharge possible.**

Concept — *Single-site gains are confined to **pain, cosmesis and stay**; operative and oncological outcomes are unchanged, and the price is lost triangulation (LESS) or longer warm ischaemia (SP).*

COMPLICATIONS OF MINIMALLY INVASIVE RENAL SURGERY

- **Complication = deviation from the normal postoperative course**; knowledge of each procedure's pitfalls prevents and manages them; **risk discussion is part of informed consent.**
- Large series: **overall complication rate about 15%**, mostly postoperative.
- **Commonest intraoperative: bleeding**, then bowel, spleen, pneumothorax, liver. **Commonest postoperative: bleeding**, wound infection, pneumonia, prolonged ileus.
- **By procedure: PN > RN > simple nephrectomy.**
- **Robotic versus laparoscopic PN: same complication risk, about half the conversion to open.**
- Patient factors: **obesity, comorbidity, prior abdominal surgery**; advanced age is not a major risk factor.
- **Prior surgery is not a contraindication**, but adhesions raise bowel injury risk → **Veress and first trocar away from scars, open (Hasson) access, or retroperitoneal approach.**
- Disease factors: **large tumours, central lesions**; conversions cluster in infective disease (chronic pyelonephritis).
- **Same-day discharge: no increase in 30-day readmission, reoperation or complications** (National Surgical Quality Improvement Program [NSQIP]).
- **Hospital volume inversely related to robotic PN morbidity, plateau at 18–20 cases per year** (may reflect surgeon volume).
- **Learning curve about 20 laparoscopic nephrectomies** — most complications in the first 20 cases; for laparoscopic PN, complications still falling beyond 750 cases.
- **Early recognition, communication and timely intervention** limit consequences; complications are shared with open surgery but **differ in type and presentation.**

Vascular Complications

- **Vascular injury is the commonest complication of minimally invasive urological surgery.**
- **Major vascular injury: <1 in 1000 — but fatal in 8–17%.**
- **Most life-threatening injuries occur at access (Veress needle, first trocar) or during hilar dissection.**
- Other causes: **excessive IVC traction, Hem-o-lok clip disconnection, Endo-GIA misfire.**
- Hilar injury: **structures divided before identification, sharp hilar dissection, stapler not covering the vessel or misfiring.**

- **Arterial injury** → **clips, sutures, cautery or stapler** as suits location and size; brisk bleeding → **hand through a lower midline incision to compress** → controlled suturing or conversion.
- **Mild-to-moderate venous bleeding** → **gauze pressure for several minutes**, dissection may continue; visible hole → **control with a grasper** → **clip or suture**.
- **Never blind clipping or suturing** — worsens injury.
- **Major injury sequence**: locate → temporary control (clamp or compression) → inform anaesthetist and theatre (blood, conversion set) → clear the field → raise pneumoperitoneum or add a trocar → expose vessel → definitive clip or suture → if not feasible, convert and call vascular surgery.
- **Inadvertent ligation** reported: IVC, aorta, contralateral renal vessels, mesenteric vessels. **Left RN**: superior mesenteric artery (SMA) mistaken for the left renal artery → **bowel necrosis**; risk higher in obesity and large medial or hilar tumours; **meticulous hilar dissection prevents it**; **open conversion for rapid reconstruction usually needed**.
- **Haemostatic device failure**: vascular staplers > non-locking titanium clips > locking clips.
- Stapler malfunction (Manufacturer and User Facility Device Experience [MAUDE] database, RN): death, conversion, bleeding, reoperation; **misfiring and jamming commonest**; **mostly operator misuse**:
 - **Firing errors**: tissue thicker than rated; stopping the firing cycle early; obstruction in the jaws.
 - **Loading errors**: improper cartridge loading; wrong-size reload.
- **Preventable**: stapling over clips; incomplete transection from malplacement → train surgeon and scrub staff.
- **End of case**: inspect; lower insufflation pressure to unmask venous bleeding; commonest sites — dissection bed, adrenal, mesentery, gonadal vessels, ureteric stump.
- **Postoperative bleeding**: hypotension + tachycardia + falling haematocrit → CT; few need exploration; renal arterial bleeding → angiography and embolisation.
- **Gross haematuria after PN** → renal artery pseudoaneurysm or arteriovenous fistula → **selective arterial embolisation (SAE)**, successful without loss of renal function; typically presents **about 12 days** after PN; **endophytic and posterior tumours at higher risk**; pseudoaneurysm the usual angiographic finding.
- **Thromboembolism**: minimally invasive approach **lowers risk versus open**; **post-discharge thromboembolism predicted by significant preoperative weight loss and chronic steroid use**; deep vein thrombosis (DVT) risk very low → **prophylaxis stratified by risk**: low risk → **sequential compression devices (SCDs)**; high risk → **pharmacological** (unfractionated or low-molecular-weight heparin); optimum controversial.

Urinary Complications

- **Persistent urine leak**: possible after any renal surgery that opens the pelvicalyceal system; falling with better renorrhaphy.
- **Incidence after laparoscopic and robotic PN 0.8–10%**; also after cyst or tumour ablation.
- **Risk factors**: large tumour, hilar location, pelvicalyceal repair, prolonged operation and warm ischaemia.
- **Management ladder**: percutaneous drainage → if it fails or distal obstruction → ureteric stent with bladder decompression, or nephrostomy.

Visceral and Bowel Complications

- **Unrecognised bowel injury**: among the most significant complications; mortality about 3%; incidence <1%.

- **Prior abdominal surgery raises risk; 32% occur at access; thermal energy near bowel causes about half of unrecognised injuries;** injury also occurs out of view during instrument passage.
- Recognised intraoperatively: **superficial thermal injury → imbricate with 3-0 silk; multiple enterotomies or devitalised segment → resection and anastomosis.**
- **Presentation: persistent, increasing pain at the trocar site nearest the injury, without erythema or pus;** distension, nausea, diarrhoea, anorexia, low-grade fever, **bowel sounds preserved, white cell count low or normal → rapid deterioration and death if missed.**
- **Thermal injuries present later than mechanical ones.**
- Fig. 142.41: day 9 after PN — distension, low-grade fever, leucopenia, pain out of proportion at one trocar site; CT dilated large bowel and free air; **caecal perforation.**
- **CT with oral contrast is the first investigation. Stable → laparoscopic re-exploration possible; unstable, distended or gross spillage → open exploration;** controlled fistula → bowel rest and parenteral nutrition (months).
- **Mesenteric defects, especially left-sided, must be closed** (internal herniation) without compromising colonic blood supply.

SOLID-ORGAN INJURY IN MINIMALLY INVASIVE RENAL SURGERY

Organ	Setting / mechanism	Management
Spleen	Left surgery (with pancreas 0–0.5%); vigorous traction on splenic ligaments	Argon beam + topical haemostat → splenorrhaphy (absorbable braided suture over bolster, or sliding clip) → splenectomy if severe
Pancreas	Exposure of left kidney, hilar dissection; insidious	Inspect at end; general surgery input; capsular → non-absorbable suture ± drain; deep → formal repair or distal pancreatectomy
Liver	Right renal surgery	Topical haemostatics + argon beam
Gallbladder	Right renal surgery	Cholecystectomy

Other Complications

- **Diaphragmatic injury** during upper-pole dissection on either side; significant injury with pneumothorax → **sudden rise in peak airway pressure, difficult ventilation** — usually recognised at once.
- **Repair sequence: purse-string suture → closed tube (back end of a closed suction drain) into the defect under suction → withdraw while tightening the purse-string during a large inspiratory breath → chest radiograph;** large or unrecognised pneumothorax → **chest drain.**
- **Unrecognised pneumothorax after laparoscopic nephrectomy about 0.3% → routine postoperative chest radiograph not recommended** in asymptomatic patients at low suspicion.
- **Neuromuscular injury in 2.7% of laparoscopic urological surgery:** abdominal wall neuralgia, motor and sensory deficit, **rhabdomyolysis**, orchialgia — from surgical trauma or positioning.
- **Rhabdomyolysis possibly commoner with laparoscopy → shorter operations; partial rather than full flank position;** suspect with **pain in back, buttocks, thighs or calves → serum creatine kinase, myoglobinuria → aggressive hydration, correct metabolic acidosis.**
- **Axillary roll in full lateral decubitus → prevents brachial plexus injury.**
- **Preserve psoas fascia in posterior dissection → prevents thigh paraesthesia.**

- Others: **surgical site infection, port-site and incisional hernia, prolonged ileus, pulmonary embolus, DVT, pneumonia.**

Concept — Minimally invasive complications are the open ones in **altered form and timing**: the lethal ones are those that happen out of view — access vessel injury, thermal bowel injury — and present late or atypically.

SUMMARY

- **MIS effectively treats surgical renal pathology**: less perioperative morbidity than open surgery, without sacrificing therapeutic outcome.

KEY POINTS: LAPAROSCOPIC AND ROBOTIC SURGERY OF THE KIDNEY

- **Less incisional pain, shorter convalescence, better cosmesis than open surgery, with comparable efficacy and long-term outcomes.**
- **Indications are the same as for open renal surgery.**
- **Oncological principles unchanged**: complete excision with negative margins, with maximal preservation of renal parenchyma where feasible.
- **Familiarity with landmarks, techniques, instruments and each procedure's pitfalls** is the key to success.